Advanced Applied Statistics
Lecture 2

Christian Hardmeier

2 September 2025

IT UNIVERSITY OF COPENHAGEN

Delta method (wrap-up)

IT UNIVERSITY OF COPENHAGEN

Central Limit Theorem

The sample mean of a large random sample of random variables with mean μ and finite variance σ^2 has approximately a normal distribution with mean μ and variance σ^2/n.

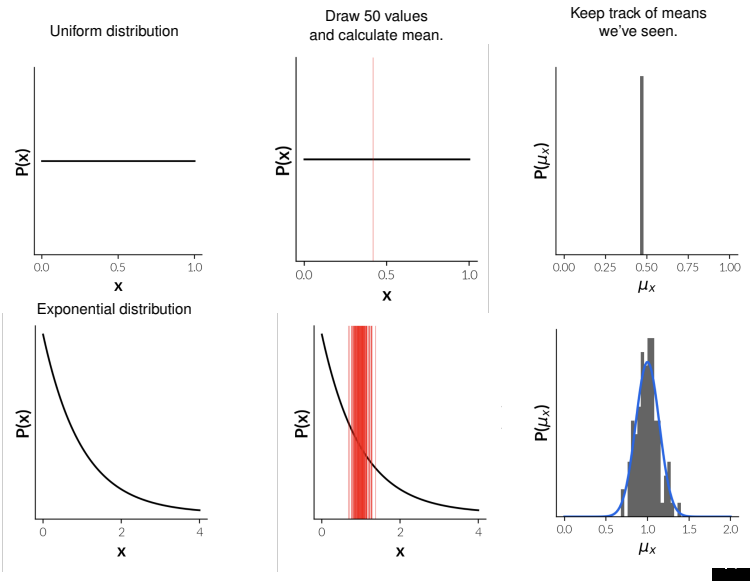
$$\sqrt{n}(X_n - \mu) \xrightarrow{d} N(0, \sigma^2),$$

That means:

- If you draw samples from **any** (sufficiently nice) distribution and calculate the mean,
- the distribution of the sample means **will be normally distributed** (even if the original distribution is not).

IT UNIVERSITY OF COPENHAGEN

Central Limit Theorem



Central Limit Theorem: Assumptions

- Random variables are *independent and identically distributed* (iid).
- Finite mean μ and variance σ^2.
- Generalisation (Liapounov):
  - Independent, but *not necessarily identically distributed* variables.
  - Additional constraint on third moment $E(X^3)$ (can't grow too fast).

Sampling distribution of precision and recall

- (Observed) precision is the mean of a set of indicator variables:

$$P = \frac{1}{N_{\text{pred } c}} \sum_{i=1}^{N_{\text{pred } c}} P_i, \qquad \text{where } P_i = \begin{cases} 1 & \text{if item } i \text{ is of class } c \\ 0 & \text{otherwise} \end{cases}$$

  and the sum is over the examples *predicted* as class c.
- If the $P_i \sim \text{Bernoulli}(p)$, then the sampling distribution will be asymptotically normal by the Central Limit Theorem:

$$P \sim N\left(p, \frac{p(1-p)}{N_{\text{pred } c}}\right)$$

- Analogous argument for recall.

F-score

- We often want to summarise Precision and Recall in a single number.
- **F-score** (or F_1) is the *harmonic mean* of Precision and Recall.

$$F = 2 \cdot \frac{P \cdot R}{P + R}$$

- If P and R are equal, then F is the same.
- If they are different, then F is closer to the *lower* of them.
- By maximising F-score, we emphasise balanced P and R.

Delta method and Taylor-series approximations

- If there is a sequence of random variable satisfying

$$\sqrt{n}(X_n - \mu) \xrightarrow{d} N(0, \sigma^2),$$

  where μ and σ^2 are finite constants and $\xrightarrow{d}$ denotes *convergence in distribution*, then

$$\sqrt{n}(g(X_n) - g(\mu)) \xrightarrow{d} N(0, \sigma^2 \cdot [g'(\mu)]^2)$$

  under mild assumptions on g.
- The first assumption is true whenever the central limit theorem holds.
- This allows us to estimate parameters of transformed quantities.

Multivariate Delta method

If we have a sequence of p-dimensional vectors $\boldsymbol{T} \in \mathbb{R}^p$ converging in distribution to $\boldsymbol{\theta} \in \mathbb{R}^p$ with asymptotic covariance matrix $\boldsymbol{\Sigma}/n$,

$$\sqrt{n}(\boldsymbol{T}_n - \boldsymbol{\theta}) \xrightarrow{d} N(\mathbf{0}, \boldsymbol{\Sigma}).$$

The Delta method approximates the variance of a transformed quantity $f(\boldsymbol{T}_i)$, $f : \mathbb{R}^p \mapsto \mathbb{R}$. For uncorrelated $\boldsymbol{T}$ (diagonal $\boldsymbol{\Sigma}$):

$$\mathrm{Var}(f(\boldsymbol{T})) \approx \frac{1}{n} \sum_{j=1}^{p} \left(\frac{\partial f}{\partial \theta_j} \right)^2 \sigma_{jj}$$

(If the elements of $\boldsymbol{T}$ are correlated, there will be additional terms involving covariances.)

Sampling distribution of F-score

- The F-score $F(P,R) = \dfrac{2PR}{P+R}$ is a transformation of the 2-dimensional statistic (P,R).

$$\mathrm{Var}(F(P,R)) \approx \frac{1}{N}\left(\left(\frac{\partial F}{\partial P}\right)^2 \mathrm{Var}(P) + \left(\frac{\partial F}{\partial R}\right)^2 \mathrm{Var}(R)\right)$$

- Assuming again $P_i \sim \mathrm{Bernoulli}(p)$ and $R_i \sim \mathrm{Bernoulli}(r)$, then

$$\mathrm{Var}(P) = p(1-p) \qquad \text{and} \qquad \mathrm{Var}(R) = r(1-r)$$

We don't know p and r, but we can use the observed P and R as plug-in estimates.

Sampling distribution of F-score

- The partial derivatives are

$$\frac{\partial F}{\partial P} = \frac{2R^2}{P+R} \qquad \text{and} \qquad \frac{\partial F}{\partial R} = \frac{2P^2}{P+R}$$

- Putting things together, we find

$$\begin{aligned}\mathrm{Var}(F(P,R)) &\approx \frac{1}{N}\left(\left(\frac{\partial F}{\partial P}\right)^2 \mathrm{Var}(P) + \left(\frac{\partial F}{\partial R}\right)^2 \mathrm{Var}(R)\right) \\ &= \frac{1}{N}\left(\left(\frac{2R^2}{P+R}\right)^2 P(1-P) + \left(\frac{2P^2}{P+R}\right)^2 R(1-R)\right)\end{aligned}$$

Fisherian Inference and Maximum Likelihood Estimation

Ronald Fisher (1908–1962)



Image source: https://www.adelaide.edu.au/library/special/exhibitions/significant-life-fisher/

Likelihood function

Probability density
$f_\mu(x) = f(x;\mu)$

Likelihood function
$L_x(\mu) = f(x;\mu)$

Log likelihood:
$$l_x(\mu) = \log L_x(\mu) = \log f_\mu(x) = \log f(x;\mu)$$

Maximum likelihood estimate $\hat{\mu}$ of μ:
$$\hat{\mu} = \arg\max_{\mu\in\Omega} l_x(\mu)$$

MLE $\hat{\theta}$ of $\theta = T(\mu)$:
$$\hat{\theta} = T(\hat{\mu})$$

Probability vs. likelihood

$$f(x;\theta) = \prod_{i=1}^{N} \theta^{x_i}(1-\theta)^{1-x_i}$$

Probability

$$f(\langle 1,1,1\rangle; 0.1) = 0.1^3 = 0.001$$
$$f(\langle 1,1,0\rangle; 0.1) = 0.1^2 \cdot 0.9 = 0.009$$
$$f(\langle 1,0,0\rangle; 0.1) = 0.1 \cdot 0.9^2 = 0.081$$
$$f(\langle 0,0,0\rangle; 0.1) = 0.9^3 = 0.729$$

Likelihood

$$f(\langle 1,1,0\rangle; 0.1) = 0.1^2 \cdot 0.9 = 0.009$$
$$f(\langle 1,1,0\rangle; 0.3) = 0.3^2 \cdot 0.7 = 0.063$$
$$f(\langle 1,1,0\rangle; 0.7) = 0.7^2 \cdot 0.3 = 0.147$$
$$f(\langle 1,1,0\rangle; 0.9) = 0.9^2 \cdot 0.1 = 0.081$$

Advantages of maximum-likelihood estimation

- *Automatic* estimate without further statistical assumptions.
- Excellent frequentist properties: Nearly unbiased in large samples.
- Bayesian justification:

$$g(\mu|x) = c_x g(\mu) \exp(l_x(\mu)) = \frac{1}{\int_\Omega g(\mu) L_x(\mu) d\mu} \cdot g(\mu) \cdot L_x(\mu)$$

  - The normalising constant/partition function c_x doesn't depend on μ.
  - If we choose a *flat* prior $g(\mu) = 1$, then neither does $g(\mu)$.
  - In that case, $g(\mu|x) \propto L_x(\mu)$.

Downsides of maximum-likelihood estimation

MLE estimates can be extremely off if estimated on little data.

- MLE estimate of the Bernoulli parameter of a coin flip based on a sample of one will always be exactly 0 or 1.

With large numbers of parameters, $\hat{\theta} = T(\hat{\mu})$ may be off even if each component of $\hat{\mu}$ is well estimated.

- Lecture 5: James-Stein estimator

Fisher Information and MLE

Score function

The *score function* is the derivative of the log-likelihood function.

$$\dot{l}_x(\theta) = \frac{\partial}{\partial\theta} \log f(x;\theta) = \frac{\dot{f}(x;\theta)}{f(x;\theta)}$$

The score function indicates how much the log-likelihood changes if you vary the parameter estimate by an infinitesimal amount, given data x.

Note: When we compute the maximum-likelihood estimate, we solve for a score of 0!

Expectation and variance of score

- Taking expectations over the density given by the *true* parameter θ_0, the score function has expectation 0.

$$\begin{aligned} E_{x\sim f(x;\theta_0)}(\dot{l}(\theta_0)) &= \int_{\mathcal{X}} \dot{l}(\theta_0) f(x;\theta_0) dx = \int_{\mathcal{X}} \frac{\dot{f}(x;\theta_0)}{f(x;\theta_0)} f(x;\theta_0) dx \\ &= \frac{\partial}{\partial\theta} \int_{\mathcal{X}} f(x;\theta_0) dx = \frac{\partial}{\partial\theta} 1 = 0 \end{aligned}$$

- If this is true at the true parameter θ_0, it will be approximately true at nearby values (continuity).
- Since the score has zero mean near θ_0, **its variance is simply** $E(\dot{l}(x)^2)$.

Fisher information

The variance of the score is called the *Fisher information*.

$$\mathcal{I}_\theta = E_{x\sim f(x;\theta)}(\dot{l}(x)^2)$$

- If the Fisher information is high, then *the MLE $\hat{\theta}$ is very sensitive to the data.*
- That means *the estimate carries a lot of information about the data* – if the data were very different, the estimate would also be very different.
- Low Fisher information means the parameter doesn't tell us much about the data.
- Larger Fisher information implies smaller variance of the MLE.
- The Fisher information is the negative curvature of the log-likelihood. High Fisher information means the log-likelihood surface is peaked.

$$E_\theta(\ddot{l}_x(\theta)) = -\mathcal{I}_\theta$$

Distribution of ML estimate

In large samples, maximum-likelihood estimates are approximately normally distributed, with a mean of θ, the true parameter, and a variance of $1/(n\mathcal{I}_\theta)$.

$$\hat{\theta} \sim N(\theta, 1/(n\mathcal{I}_\theta))$$

For iid normal data, $x_i \sim N(\theta, \sigma^2)$,

$$\hat{\theta} \sim N(\theta, \sigma^2/n)$$

A key principle of *Fisherian inference* is to bring estimates into a form where we can reduce complex inferential questions to inference about normal distributions.

Application: Hypothesis test for a Laplace model

How can we test a hypothesis about an estimated parameter?

- Assume we have a Laplace model:

$$f(x|\theta) = \frac{1}{2b} \exp\left(-\frac{|x-\theta|}{b}\right)$$

- We'd like to test the null hypothesis $H_0 : \theta = \theta_0$.
- The exact sampling distribution of θ is difficult to work with.
- It can be shown that the ML estimator of the data is the sample median, and the Fisher information is $\mathcal{I}_\theta = b^{-2}$.
- The MLE is distributed approximately as

$$\hat{\theta} \sim N(\theta, 1/(n\mathcal{I}_\theta)) = N(\theta, b^2/n).$$

- For $n = 100$, $b = 1$ and under the null hypothesis H_0, we expect 95% of the estimates to be within

$$\hat{\theta} \in (\theta_0 - 0.196, \theta_0 + 0.196).$$

(Ly et al., *J. Mathematical Psychology* 80 (2017) 40–55, Example 4)

Application: Study design

How large a sample do we need to ensure that the MLE is probably in the right ballpark?

- Assume: Bernoulli (coin flip) model with parameter $\theta = \Pr(X = 1)$.
- We'd like the estimate $\hat{\theta}$ to be within 0.1 of the true value θ^* with 68% probability ($\pm$ 1 standard deviation).
- The MLE has a normal sampling distribution, $\hat{\theta} \sim N(\theta, 1/(n\mathcal{I}_\theta))$, so

$$\sigma(\theta) = \frac{1}{\sqrt{n\mathcal{I}_\theta}}$$

- We don't know θ^*, so we allow for the worst case.

(Ly et al., *J. Mathematical Psychology* 80 (2017) 40–55, Example 2)

Application: Study design

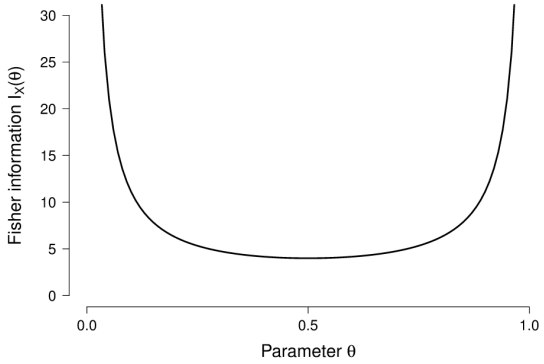


Fig. 1. The unit Fisher information $I_X(\theta) = \frac{1}{\theta(1-\theta)}$ as a function of θ within the Bernoulli model. As θ reaches zero or one the expected information goes to infinity.

(Ly et al., 2017)

Application: Study design

- The worst case (minimum information) is $\theta = 0.5$, where

$$\mathcal{I}_\theta = \frac{1}{\theta(1-\theta)} = 4.$$

- In this case,

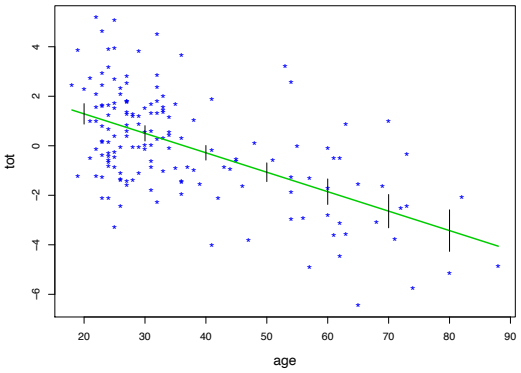
$$\sigma(0.5) = \frac{1}{\sqrt{n\mathcal{I}_\theta}} = \frac{1}{2\sqrt{n}}.$$

- We want $\sigma = \frac{1}{2\sqrt{n}} \leq 0.1$. Solving for n, we find we must have

$$n \geq 25.$$

Generalised Linear Models

Linear regression



(Fig. 1.1 in CASI)

$$y \sim N(\alpha_0 + \alpha_1 x, \sigma^2)$$

- Data assumed normally distributed (normal residuals).
- Mean predicted by regression line.
- Variance assumed constant.
- But what if data are not normally distributed?

Non-normal quantities of interest

- Counts (non-negative integers)
  Poisson regression
- Proportions (non-negative reals between 0 and 1)
  Logistic regression
- Waiting times (non-negative reals)
  Gamma regression
- Ratios (often heavy-tailed distributions)
  ???
- ...

Exponential families

An exponential family is a parametric family of distributions that can be expressed in the following form:

$$f_\lambda(x) = e^{\lambda y - \gamma(\lambda)} g_{\lambda_0}(x)$$

- λ: *Natural* or *canonical parameter*.
  Not necessarily the "usual" parameter.
- $y = t(x)$: *Sufficient statistic* with fixed dimensionality (regardless of data size).
  Depends on x, but not on λ.
- $f_{\lambda_0}(x)$: *Carrying density*, which defines the shape of the density.
  Depends on x, but not on λ. (λ_0 is a constant.)
- $\gamma(\lambda)$: *Normalising function*, which scales the density to integrate (or sum) to 1.
  Depends on λ, but not on x.

Sufficient statistic

- A statistic $t = t(X)$ is *sufficient* with respect to a statistical model and its associated parameter θ if "no other statistic that can be calculated from the same sample provides any additional information as to the value of the parameter" (Fisher, 1922).
- The conditional distribution of the data X given the statistic $t(X)$ does not depend on the parameter.

$$\Pr(X|\theta, t(X)) = \Pr(X|t(X))$$

- Exponential families are specials in that they admit sufficient statistics of fixed dimension.
  - Sample mean and sample variance are sufficient for an iid normally distributed sample.

Example: Poisson distribution

Express $f_\theta(x) = \dfrac{\theta^x e^{-\theta}}{x!}$ in the exponential family form $f_\lambda(x) = e^{\lambda y - \gamma(\lambda)} g_{\lambda_0}(x)$.

- Take logs:

$$\begin{aligned}\log f_\theta(x) &= \log \theta^x + \log e^{-\theta} - \log x! \\ &= x \log \theta - \theta - \log x!\end{aligned}$$

- The only term with both x and θ is $x \log \theta$.
  To convert this into the form λy, set $\lambda = \log \theta$ and $y = x$.
- This leads to $\gamma(\lambda) = e^\lambda$ and $g_{\lambda_0}(x) = \dfrac{1}{x!}$.

Generalised linear regression

Generalised linear models are a principled way to apply regression to quantities that are not normally distributed.

- Data are from a one-parameter exponential family.
- *Natural parameter* modelled as an affine function of the features.
- The means are connected to the natural parameter by a *link function* that depends on the assumed distribution.

$g(x) = \alpha_0 + \alpha_1 x$	or	$E(Y\|X) = g^{-1}(\alpha_0 + \alpha_1 x)$	(univariate case)
$g(x) = \mathbf{X}\alpha$	or	$E(Y\|X) = g^{-1}(\mathbf{X}\alpha)$	(general case)

Logistic regression

- Logistic regression models proportions of counts.
- The underlying exponential family is the *binomial distribution*.
  Binomial distribution: Probability of k successes in n trials,
  given an independent success probability of π.
- The natural parameter is the *logit*:

$$\lambda = \log \frac{\pi}{1-\pi} \qquad (\lambda \in \mathbb{R})$$

Let $\{f_\lambda(y) = \exp(\lambda y - \gamma(\lambda)) f_0(y)\}$ be a one-parameter family of densities.
Let $\mathbf{X} \in \mathbb{R}^{N \times p}$ be the *design matrix* with N items described by p features, $\alpha \in \mathbb{R}^p$ the regression weights, and $\mathbf{y} = (y_1, y_2, \ldots, y_N)$ the vector of responses.
Regression assumptions:

$$y_i \sim f_{\lambda_i}(x) \text{ independently, and } \lambda = \mathbf{X}\alpha.$$

Now

$$f_\alpha(\mathbf{y}) = \prod_{i=1}^N f_{\lambda_i}(y_i) = \exp\left(\sum_{i=1}^N (\lambda_i y_i - \gamma(\lambda_i))\right) \prod_{i=1}^N f_0(y_i) = e^{\alpha^\top \mathbf{z} - \psi(\alpha)} f_0(\mathbf{y})$$

$$\text{where } \mathbf{z} = \mathbf{X}^\top \mathbf{y} \text{ and } \psi(\alpha) = \sum_{i=1}^N \gamma(\mathbf{x}_i^\top \alpha).$$

This is a p-parameter exponential family
with natural parameter $\alpha \in \mathbb{R}^p$ and sufficient statistic $\mathbf{z} \in \mathbb{R}^p$.

Advantages of GLMs with canonical link

- Canonical link means $g(x)$ maps from the *natural parameter* to the mean.
- Score function is linear in the sufficient statistic.
- Likelihood function is concave.
- Natural parameter ranges over the real line.
  Mean may have constrained range (e.g., between 0 and 1 in logistic regression).
- Other link functions *can* be used.

Regression: Take-aways

- Regression reduces prediction variance for one instance by exploiting information from other instances.
- Generalised linear models are useful for regression when the data distribution is not normal.
- Exponential families provide a consistent theory for many of the most common distributions. Particularly useful:
  - Logistic regression for proportions (or classification).
  - Poisson regression for counts.
- Exponential families admit fixed-dimensional sufficient statistics.